

JUAN SEBASTIAN ZAPATA ACOSTA

Bogotá, Colombia

Microbiologist | MSc Candidate in Computational Biology | Bioinformatics & Data Science

+573232843437 | 10.sebastian.zapa@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Microbiologist and MSc candidate in Computational Biology with experience spanning microbiology, bioinformatics, biological data analysis, metagenomics, and laboratory operations management. Experienced in developing and applying computational workflows for microbial community analysis, functional annotation, metagenome assembly, and predictive biological modeling. Skilled in R, Python, statistical analysis, and scientific data visualization, with demonstrated experience leading laboratory teams, managing quality systems, and translating complex biological datasets into actionable insights for research and industrial applications.

PROFESSIONAL EXPERIENCE

FLORES SAGARO S.A.

Microbiology & Bioinput Development Coordinator | 2025 – Present

- Coordinated microbiological workflows and operational quality-related processes.
- Managed experimental records, microbiological monitoring data, and technical documentation.
- Supported process consistency, contamination control practices, and operational reporting activities.
- Performed data analysis and process optimization based on microbiological and production-related information.
- Collaborated across operational and technical teams to maintain standardized procedures and workflow efficiency.

TECMOL PHARMACEUTICALS

Microbiology Analyst | March 2025 – April 2025

- Conducted microbiological quality assessments for pharmaceutical, agronomic, and veterinary products.
- Performed microbiological testing and analysis following USP pharmacopeia standards and standardized laboratory procedures.
- Supported quality assurance activities through documentation review, data recording, and compliance-oriented workflows.
- Contributed to contamination control and microbiological monitoring processes to ensure product safety and quality standards.

BBC BREWERY DE LA SABANA S.A.S.

Microbiologist – Quality Analyst | October 2024 – February 2025

- Performed microbiological monitoring across multiple production stages including fermentation, malting, packaging, and finished product evaluation.
- Conducted microbial enumeration and quality control testing to support process consistency and product safety.

- Maintained technical documentation, traceability records, and quality-related reports under standardized operational procedures.
- Supported deviation identification and process monitoring activities within a quality-focused industrial environment.

CONCENTRIX

Bilingual Customer Support Agent - Bogotá, Colombia | January 2024 – June 2024

- Delivered customer support in English and Spanish within an international environment.
- Managed documentation, issue tracking, and workflow organization.
- Worked under KPI-based systems and performance metrics.
- Developed strong communication, organization, and remote collaboration skills.

UNIVERSITY OF LOS ANDES – BIOPHYSICS INVESTIGATION GROUP

Research Assistant | 2021 – 2023

- Processed and analyzed biological datasets using bioinformatics and computational tools.
- Supported experimental data interpretation through data processing and analytical workflows.
- Worked with biological samples, laboratory instrumentation, and interdisciplinary research methodologies.
- Assisted in organizing, validating, and documenting research data for scientific projects
- Collaborated with research teams integrating laboratory and computational approaches.

PROJECTS

Metagenomic Characterization of Vermicomposting Microbiomes

Master's Research Project

- Reconstruction of microbial genomes from shotgun metagenomic data.
- Functional characterization of microbial communities.
- Evaluation of microbial ecological interactions.
- Development of computational workflows for biological data analysis.

Image Processing and Computational Analysis

GitHub Project

- Scientific image processing workflows using Python.
- Data extraction and quantitative analysis.
- Computational tool development for biological applications.

EDUCATION

UNIVERSIDAD DE LOS ANDES
MSc in Computational Biology (Ongoing) |
Bogotá, Colombia

UNIVERSIDAD DE LOS ANDES
BSc in Microbiology | Bogotá, Colombia

- Metagenomics
- Bioinformatics
- Microbiome Analysis
- Functional Annotation
- MAG Recovery
- Omics Data Analysis
- Biological Data Integration

Data Science

COMPETENCIES

- Python
- R

- Machine Learning
- Statistical Modeling
- Multivariate Analysis
- Data Visualization

Computational Biology

- Genome-Scale Metabolic Models (GEMs)
- Microbial Ecology
- Community Modeling
- Biological Network Analysis
- Predictive Analytics

Leadership & Operations

- Team Leadership
- Laboratory Management
- Quality Control
- Scientific Reporting
- Process Optimization

TECHNICAL SKILLS

Programming

- Python
- R

R Ecosystem

- ggplot2
- phyloseq
- dplyr
- tidyr
- vegan
- microbiome analysis workflows

Bioinformatics Software

Pre-processing

- Fastp
- Bowtie2
- Samtools

Metagenome Assembly

- Megahit
- metaSPAdes

Binning & MAG Recovery

- MetaBAT2
- SemiBin2
- DAS Tool

Functional Annotation

- Bakta
- metabolic
- Funannotate

Model Construction

- CarveMe
- CarveFungi
- CheckM2
- KBase

Community Simulation

- MICOM
- COMETS
- CometsPy
- Gurobi
- GLPK

Data Analysis

- PCA
- PCoA
- PERMANOVA
- Alpha Diversity
- Beta Diversity
- Functional Profiling

LANGUAGES

- Spanish — Native
- English — B2 (Professional Proficiency)
- Portuguese — Basic
- German — Basic